FIN42350 Capital Markets - Group Project

A Systematic Framework for Client-Specific Portfolio Construction and Risk Analysis

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Abstract

This report develops a systematic framework for constructing a multi-asset portfolio based on an investor specific preference. The frame has been designed into five phases using in-sample historical data for portfolio allocation and out-of-sample data for the optimised portfolio performance evaluation. In addition, we use Monte Carlo simulation future expected return under three different investor profiles, risky, balanced, and conservative with SPY as a proxy for market benchmark.

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1 Introduction

We present an end-to-end systematic framework for multi-asset portfolio design and construction. Conceptually, we distinguish the design process into two decision layers. The first layer determine asset allocation based on investor risk aversion profile score, while second layer optimise securities allocation within each asset class.

We begin with an overview of Portfolio Design and Construction Process, describing the end-to-end approch that has been designed. The framework comprises into five main phases. Phase I — Investor Profile and Risk Assessment, examing main investor questionnaire, the importance assigned to each question as weight, and calculation of risk score. Phase II — Investor Suitability and Asset-Class Allocation, we discuss the first decision layer of main asset class allication based on predified asset suitability lookup approach. Phase III — Market Data Preparation, we examin dataset construction, including currency conversion, annualised return calculation, and muti-assets variance/co-variance matrix creation. Phase IV — Portfolio Optimisation combines historical data preparation, investor risk aversion coefficient, asset class allocations proportions and optimise the exact allication percentage wihin each of the the asset class instruments for optimial performance. Finally, Phase V — Portfolio Benchmarking and Performance Analysis, evaluate the proposed portfolio using out-of-sample period, an EUR-denominated S&P 500 benchmark, Value at Risk (VaR) and Expected Shortfall (ES), and Monte Carlo simulation. The simulation estimates the distribution of possible outome under physical market measurement.

The Concrete Client Run and Results, demonstrates how the framework converts investor information and historical market data into an implementable portfolio recommendation.

The report concludes with the Conclusion, which summarises the principal findings, limitations, and final portfolio recommendation.

2 Portfolio Design and Construction Process

In this section we give an overview of the design and the main principles was held during this process. Figure 1 shows the end-to-end design steps.

Phase I — Investor Profile and Risk Assessment step begins with a set of questions designed to capture investor-specific characteristics. The investor's answers are then weighted according to the importance assigned to each question by the investor preference. The principle rationale for such decision, given two different investors profiles who provided identical responses but place different levels of importance on each question would recieve different risk scores and profile recommendations. Finally, the investor risk score is mapped to a risk-aversion coefficient, λ, which serves as an input to the portfolio-weight optimisation in Phase IV.

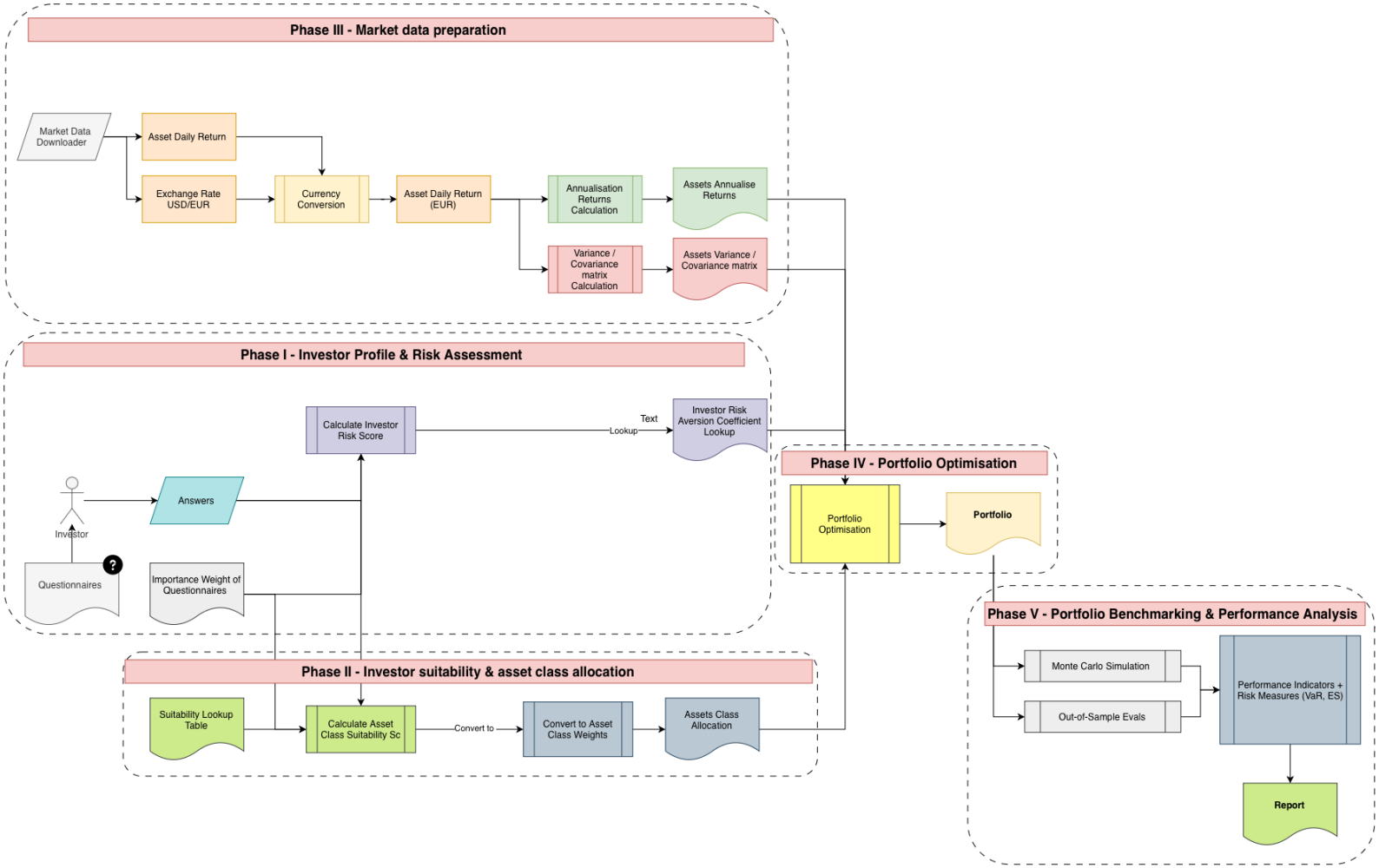


Figure 1: End-to-end portfolio construction and evaluation workflow

Phase II — Investor Suitability and Asset-Class Allocation step main objective is to provide an allocation for each asset class basded on asset-class suitability score S_c. Calculating asset-class suitability score S_c depneds on (i) Investor's questionnaire importance weight (ii) Asset class suitability lookup table. The output of this step will be feeding as a constraint into Phase IV - Portfolio Optimisation.

Phase III — Market Data Preparation step focus on retrieval financial market data from available data source, and adjusting the frequency of return, in our case daily we utilise Yahoo Finance and adjust for daily return. In addition, we retrieved USD/EUR exchange rate, and applying rate conversion on price action before return/annualised returns calculations. Finally, we construct variance-covariance matrix which serves as an input to the portfolio-weight optimisation in Phase IV.

Phase IV — Portfolio Optimisation step taking in calcuated variance-covariance metix Σ, asset class allocation vector $\vec{W}$, and investor risk aversion coefficient and optimise against allocation within asset class, producing taregt portfolio.

Phase V — Portfolio Benchmarking and Performance Analysis final phase focus is to conduct full assessment of the resultant portfolio. The assesment start with out-of-sample evaluation of 2-years, following a calcuation of set of performance indicators including, and finally we apply Monte Carlo simulation to produce expected potensional return given allocated assets expected return and variaiance.

In the following next sections, we discuss in-deepth each of the five phases and provide sample output as examples.

3 Phase I — Investor Profile and Risk Assessment

We start by building the investor's profile and quantifying their risk aversion through risk aversion coefficient.

3.1 Investor questionnaire and weights

The investor is asked the questions presented in Table 1. The discretisation of year ranges make the investment-horizon choices reproducible.

Table 1: Investor questionnaire and five-level answer scales

Attribute	Question presented to the investor	Five-level response range
Risk tolerance	How much short-term loss could you tolerate without selling your investments?	Very low–Very high (5 levels)
Investment horizon	When do you expect to begin using most of this investment?	Very short (<1 year); Short (1–3 years); Medium (3–5 years); Long (5–10 years); Very long (>10 years)
Growth objective	How strongly do you prioritise long-term capital growth over current income?	Very low–Very high (5 levels)
Capital preservation	How important is avoiding a loss of the amount initially invested?	Very low–Very high (5 levels)
Liquidity requirement	How much of this portfolio might you need to withdraw at short notice?	Very low–Very high (5 levels)

We attached an importance points for each of the Attribute defined, where the total sum of all

the weight would be summed to 10, and thier normalised values summed equal to one.

Table 2 provides an illustrative set of responses and investor-assigned importance points. The points sum to 10 and are normalised to weights that sum to one.

Table 2: Illustrative questionnaire responses, importance weights, and answer normalisation

Attribute and key	Example answer	Importance points	Normalised weight	Values mapped from 0 to 1	Direction
0. Risk tolerance `risk_tolerance`	Moderate	3	0.30	Very low = 0; Low = 0.25; Moderate = 0.50; High = 0.75; Very high = 1	Increasing
1. Investment horizon `investment_horizon`	Long	2	0.20	Very short = 0; Short = 0.25; Medium = 0.50; Long = 0.75; Very long = 1	Increasing
2. Growth objective `growth_objective`	High	2	0.20	Very low = 0; Low = 0.25; Medium = 0.50; High = 0.75; Very high = 1	Increasing
3. Capital preservation `capital_preservation`	Medium	2	0.20	Very low = 1; Low = 0.75; Medium = 0.50; High = 0.25; Very high = 0	Decreasing
4. Liquidity requirement `liquidity_requirement`	Low	1	0.10	Very low = 1; Low = 0.75; Medium = 0.50; High = 0.25; Very high = 0	Decreasing
	Total	10	1.00		

3.2 Investor risk score and risk-aversion coefficient

The normalised investor risk score $R_Q \in [0, 1]$ quantifies the investor's risk profile and is mapped to the risk-aversion coefficient λ, as shown in Table 3.

$$R_Q = \sum_{j=1}^{J} \alpha_j q_j, \qquad \sum_{j=1}^{J} \alpha_j = 1, \tag{1}$$

where $q_j \in [0, 1]$ is the normalised answer and α_j is the question's importance weight.

Table 3: Investor risk-score intervals, profiles, and risk-aversion coefficients

R_Q interval	Profile	λ
$[0.00, 0.20)$	Very conservative	12
$[0.20, 0.40)$	Conservative	8
$[0.40, 0.60)$	Moderate	5
$[0.60, 0.80)$	Growth	3
$[0.80, 1.00]$	Aggressive	1

Risk-aversion coefficient λ, is the used as a control variable of how strongly the portfolio optimiser penalise risk relative to expected return in Phase IV.

4 Phase II — Investor Suitability and Asset-Class Allocation

The first decision layer, after we collect investor's questionnaire answers and importance points, would be to perform asset class allocation. Two components are needed (i) Investor's questionnaire importance weight α_j, (ii) Asset-suitability lookup $A_{j,k_j,c} \in [0, 1]$.

4.1 Asset suitability lookup

The asset-suitability lookup is a predefined scoring table that translates an investor's questionnaire answers into suitability scores for each asset class.

For every combination of:

$$\text{Question} \times \text{Answer} \times \text{Asset class},$$

the lookup assigns a score:

$$A_{j,k,c} \in [0, 1],$$

where:

- j identifies the questionnaire question;
- k is the investor's selected answer;
- c identifies an asset class;
- $A_{j,k,c}$ is the suitability score

And 0 means unsuitable and 1 means highly suitable.

For example, the complete risk-tolerance slice is shown in Table 4.

Table 4: Risk-tolerance slice of the asset-suitability lookup

Risk tolerance	Equity	Fixed income	Cash	Real estate	Commodity
Very low	0.10	0.80	1.00	0.30	0.20
Low	0.30	0.90	0.90	0.40	0.30
Moderate	0.60	0.80	0.60	0.60	0.50
High	0.85	0.50	0.30	0.75	0.70
Very high	1.00	0.30	0.10	0.85	0.85

4.2 Weighted asset-class suitability

Arriving to the asset class weights that satisfies investor profile would require adjusting selected suitability score by investor answer's importance points for each asset class S_c.

$$S_c = \sum_{j=1}^{J} \alpha_j A_{j,k_j,c}. \tag{2}$$

where, α_j represents investoer's answer importance points, and $A_{j,k_j,c}$ representation asset-suitability lookup for question j, answer k, and asset class c.

Table 5 presents an example of the resulting suitability score for each asset class.

Table 5: Illustrative asset-class suitability scores

No.	Asset class	Suitability score
0	Equity	0.730
1	Fixed income	0.705
2	Cash	0.545
3	Real estate	0.630
4	Commodity	0.550

4.3 Conversion to asset-class weights

In the final stage of the first decision layer, each asset-class suitability score, S_c, is normalised to obtain its corresponding strategic asset-class weight, W_c:

$$W_c = \frac{S_c}{\sum_{c \in \mathcal{C}} S_c}, \qquad \sum_{c \in \mathcal{C}} W_c = 1, \tag{3}$$

where $\mathcal{C}$ denotes the set of eligible asset classes. This normalisation converts the suitability scores into proportional allocations that sum to one.

The resulting asset-class weight vector, $\mathbf{W}$, is subsequently imposed as a set of constraints in the second decision layer, in which the optimiser determines the allocation among the eligible securities within each asset class.

5 Phase III — Market Data Preparation

5.1 Asset universe and market-data download

The investable universe contains 134 instruments spanning nine asset classes, 78 asset subclasses, and 31 geographic classifications. Table 6 show a head and tail sample of downloaded USD/EUR exchange rates. Table 7 summarises its composition, while Appendix A reports the complete instrument list. Full List of Asset universe are in Appendix A.

Table 6: Sample of the daily USD/EUR exchange-rate series

Observation	Date	USD/EUR rate
0	2019-10-03	0.91210
1	2019-10-04	0.91113
2	2019-10-05	0.91113
3	2019-10-06	0.91113
4	2019-10-07	0.91020
	...	
2553	2026-09-29	0.87930
2554	2026-09-30	0.88177
2555	2026-10-01	0.88283
2556	2026-10-02	0.88830
2557	2026-10-03	0.88830

Table 7: Summary of the investable asset universe

Asset class	Instruments	Subclasses	Geographies
Equity	81	39	30
Fixed income	19	11	3
Commodity	15	12	1
Currency	7	7	7
Volatility	5	4	1
Real estate	3	1	1
Crypto asset	2	2	1
Cash	1	1	1
Alternatives	1	1	1
Total	134	78	31

Yahoo Finance API are used as an endpoint to download market data. Total of 7 years of data were downloaded, with a chronological 5-years/2-years split for an in/out-of sample respectively.

Table 8: Chronological in-sample and out-of-sample split

Dataset	Period	Purpose
Full sample	3 Oct 2019–3 Oct 2026	Full dataset (7 years)
In-sample	3 Oct 2019–2 Oct 2024	Portfolio design and construction (5 years)
Out-of-sample	3 Oct 2024–3 Oct 2026	Evaluate the frozen portfolio (2 years)

5.2 Currency conversion and EUR daily returns

The exchange rate of USD/EUR were downloaded for the full period 7-years i.e. in/out-of sample, and applied conversion before calcuating daily return.

For a USD-denominated asset, we apply:

$$P_{i,t}^{\mathrm{EUR}} = P_{i,t}^{\mathrm{USD}} X_t^{\mathrm{USD/EUR}}, \qquad r_{i,t}^{\mathrm{EUR}} = \frac{P_{i,t}^{\mathrm{EUR}}}{P_{i,t-1}^{\mathrm{EUR}}} - 1. \tag{4}$$

5.3 Annualised return estimates

Following daily return calcuation in EUR, annulaised return are calcuated to express an investement's return on a one-year bases. This would facilitate assessing level of investment horizon especially for longer term period.

$$\hat{\mu}_i = 252\, \bar{r}_{i,d}. \tag{5}$$

5.4 Variance and covariance estimates

A variance-covariance matrix is calcuated out of annualised return $\hat{\Sigma}$, where it will be used as part of asset allocation with class in Phase IV.

$$\hat{\Sigma} = 252 \ \mathrm{Cov}(r_d), \qquad \hat{\sigma}_i = \sqrt{\hat{\Sigma}_{ii}}. \tag{6}$$

6 Phase IV — Portfolio Optimisation

Phase IV represents the second decision layer of the portfolio design framework. In this phase, the optimiser determines how each assets are allocated within each asset class.

6.1 Optimiser inputs

The optimiser combines five inputs produced by the preceding phases:

1. the annualised expected EUR return vector, $\hat{\boldsymbol{\mu}}$;
2. the annualised EUR return covariance matrix, $\hat{\boldsymbol{\Sigma}}$;
3. the investor's risk-aversion coefficient, λ;
4. the required asset-class weights, W_c; and
5. the set of eligible ETFs within each asset class.

6.2 Mean–variance objective

The optimiser objective function select ETF weights that maximise the portfolio's expected return penalising overall variance. Using investoer's risk aversion coefficient λ as a controlling coefficient.

A high investor's risk available coefficient λ would result in greater emphasis on reducing the overall portfolio variance and therefore favours less risky allocation, and true vice versa.

$$\max_{w} \left(w^{\top} \hat{\mu} - \frac{\lambda}{2} w^{\top} \hat{\Sigma} w \right). \tag{7}$$

6.3 Portfolio constraints

$$\sum_{i} w_i = 1, \tag{8}$$

$$w_i \geq 0, \tag{9}$$

$$\sum_{i \in c} w_i = W_c, \tag{10}$$

$$w_i \leq u_i. \tag{11}$$

The portfolio is subject to four principal constraints:

First, short selling is prohibited, so each ETF weight must be non-negative. Second, the summation of all ETFs within same asset class should be equal to the pre-defined asset class allocated weight W_c, that was defined in the first layer decision. Third, the profile must be fully invested. Finally, an upper limit is imposed to reduce concentration risk.

6.4 Portfolio output and validation

The in sample optimisation yield a fully investment portfolio with asset being allocated based on investor's risk aversion preference. Example of such allocation in Table 9.

Table 9: Sample optimised portfolio allocation

Display name	Investment type	Geography	Asset type	Allocation (%)	Allocation (€)
SPDR Bloomberg 1–3 Month T-Bill ETF	ETF	United States	Treasury bills	17.25	172,468.35
VanEck Semiconductor ETF	ETF	Global	Semiconductors	15.00	150,000.00
Schwab US TIPS ETF	ETF	United States	Inflation-linked bonds	15.00	150,000.00
Real Estate Select Sector SPDR Fund	ETF	United States	Listed real estate	15.00	150,000.00
United States Brent Oil Fund	ETF	Global	Brent crude oil	7.34	73,444.30
iShares TIPS Bond ETF	ETF	United States	Inflation-linked bonds	7.31	73,101.27
iShares Semiconductor ETF	ETF	United States	Semiconductors	4.94	49,421.29
Vanguard Real Estate ETF	ETF	United States	Listed real estate	4.94	49,367.09
iShares Silver Trust	ETF	Global	Silver	4.93	49,342.22
Global X MSCI Argentina ETF	ETF	Argentina	Country equity	3.16	31,591.37
Invesco DB Agriculture Fund	ETF	Global	Agriculture	2.81	28,055.50
iShares Gold Trust	ETF	Global	Gold	1.72	17,227.97
United States Copper Index Fund	ETF	Global	Copper	0.60	5,980.65
Total				100.00	1,000,000.00

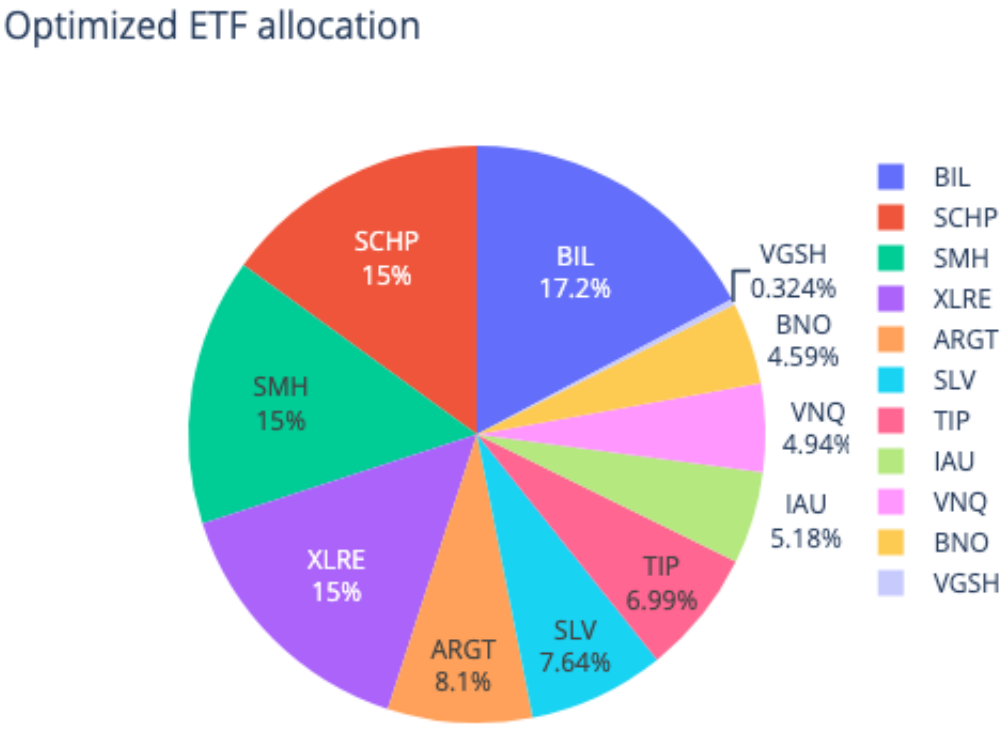


Figure 2: Sample optimised allocation of the €1 million portfolio

7 Phase V — Portfolio Benchmarking and Performance Analysis

The final phase perform set of evaluations on the optimised portfolio, we start with benhmark calcification with out-of-sample of 2-years using S&P 500 as a proxy to the market. Following we calculate Value at Risk (VaR) and Expected Shortfall (ES) at the 95% and 99% confidence levels. Finally, we apply Monte Carlo simulation to produce expected potensional return given allocated

assets expected return and variaiance.

7.1 Benchmark and out-of-sample evaluation

The optimised portfolio weights were estimated exclusive in-sample data. subsequently, we apply benchmark on the out-of-sample portion of the dataset measuring the overall performance for the 2-years without rebancing.

An initial capital of €1 million would be allocated based on the produced weights, following by daily return calculating for each trading day based on the adjusted closing price. Since both the portfolio and benchmark contain US-dollar-denominated securities, their adjusted prices were converted into euros using the corresponding daily USD/EUR exchange rate, which was handled in Phase III - Market Data Prepration.

SPY was selected as the benchmark reflecting the market as a whole. The cumulative value of the portfolio and benchmark was calculated from the same initial investment:

$$V_t = V_0 \prod_{\tau=1}^{t} (1 + r_\tau),$$

where V_0 is the initial capital and r_τ is the daily euro-denominated return. A visualisation of the benchmark is shown in Figure 3.

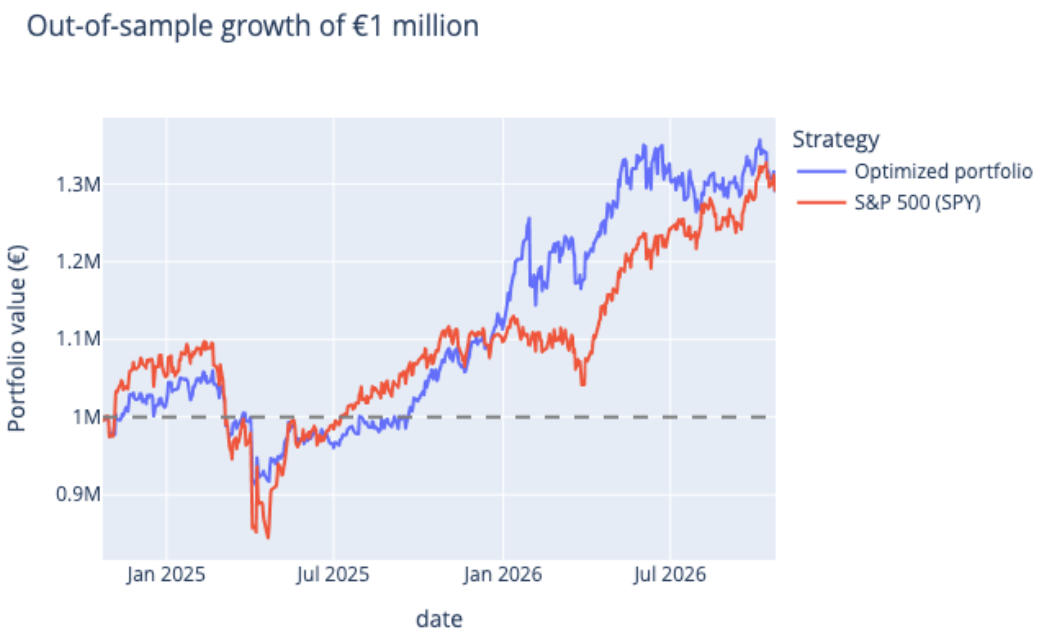


Figure 3: Out-of-sample growth of the optimised portfolio and S&P 500 benchmark from an initial investment of €1 million

7.2 Performance indicators

Portfolio performance was evaluated starting with arithmatic average return, geometic average return, Risk-adjusted performance was assessed using the Sharpe ratio, which relates excess

return to total volatility, and the Treynor ratio, which relates excess return to systematic risk measured by beta. Jensen's alpha was used to identify returns exceeding those predicted by the CAPM, given the portfolio's market exposure. The information ratio measures alpha relative to residual risk, or tracking error, while the M^2 measure expresses Sharpe-based performance as an easily interpreted differential return relative to the market benchmark. Beta and R^2, estimated against SPY, provide supporting evidence concerning the portfolio's systematic market exposure and the proportion of its return variation explained by the benchmark. As an exmaple Table 10 shows an outcome of the highlighted indicators relative the the S&P 500 benchmark.

Table 10: Example mean simulated performance indicators

Performance indicator	Optimised portfolio	S&P 500 benchmark
Arithmetic annualised return	14.13%	17.41%
Geometric annualised return	15.30%	19.02%
Annualised volatility	16.24%	22.03%
Sharpe ratio	0.834	0.790
Beta versus S&P 500	0.661	1.000
Jensen's alpha	2.01%	0.00%
Treynor ratio	0.204	0.174
Residual risk	7.19%	0.00%
Information ratio	0.262	N/A
M^2	0.98%	0.00%
R^2	80.19%	100.00%

7.3 Value at Risk and Expected Shortfall

Downside risk was examined using historical Value at Risk (VaR) and Expected Shortfall (ES) at the 95% and 99% confidence levels respectively. VaR estimates the loss threshold exceeded with a specified probability, whereas ES measures the average loss conditional on returns falling beyond that threshold. These two risk measures were conducted using the out-of-sample euro denominated SPY returns. Figure 4 shows an out-of-sample daily loss distribution with 95% and 99% Value at Risk thresholds.

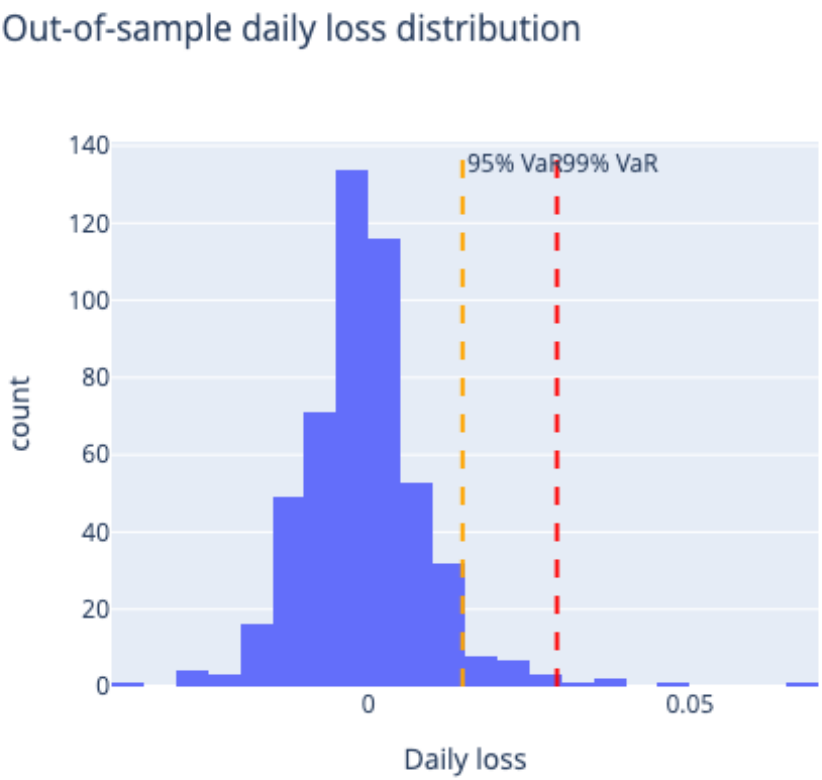


Figure 4: Out-of-sample daily loss distribution with 95% and 99% Value at Risk thresholds

7.4 Monte Carlo simulation

Finally, a forecast-based evaluation apprach were used to asset expected return using Monte Carlo simulation. We simulated asset returns using in-sample annulaised expected return and covariance matrix, to preserve dependency between the portfolio assets across 252 trading days.

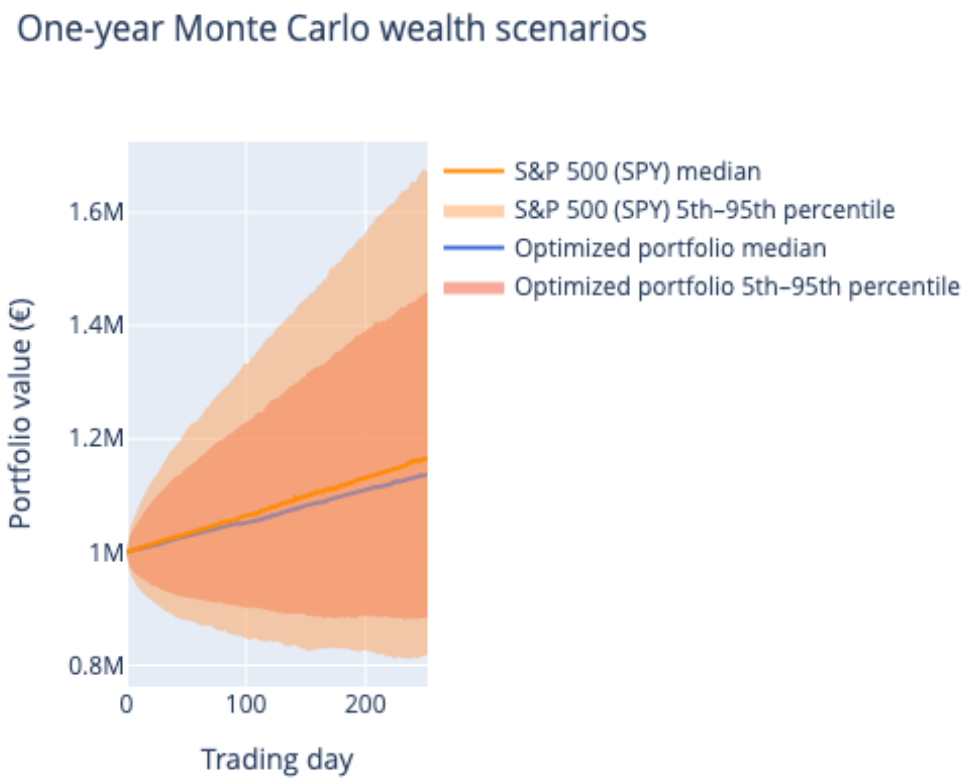


Figure 5: One-year Monte Carlo wealth scenarios for the optimised portfolio and S&P 500 benchmark

8 Three Investor Scenarios and Results

8.1 Scenario definitions and common assumptions

We perfomed senstivity analysis using three investor scenarios, demonstrating different views from risk point of view. (i) Risky investor emphasis growth over value, (ii) balanced investor combine growth and capital preservation, and conservative investor prioritise capital preservation and liquity over growth. Each scenario begins with €1 million and uses the same asset universe, euro-denominated market data, estimation window, portfolio constraints methodology, and S&P 500 benchmark. Table 11 showes the three investor outlook against the five questionnaire attribute.

Table 11: Investor-scenario questionnaire responses

Questionnaire attribute	Risky	Balanced	Conservative
Risk tolerance	Very high	Moderate	Low
Investment horizon	Very long	Medium	Short
Growth objective	Very high	Medium	Low
Capital preservation	Low	Medium	High
Liquidity requirement	Low	Medium	High

8.2 Phase I results: investor risk assessment

The outcome of the first phase, highlight different investor risk score with the highest value attributed to the most risky invesotr portofilio and λ of 1, and progress accordingly for balanced and conservative investors, as shown in Table 12.

Table 12: Investor risk results by scenario

Scenario	Risk score R_Q	Risk profile	λ
Risky investor	0.950	Aggressive	1
Balanced investor	0.500	Moderate	5
Conservative investor	0.250	Conservative	8

8.3 Phase II results: suitability and asset-class allocation

In phase II, the three sets of suitability scores are normalised separately across the available asset classes. This produces scenario-specific strategic allocations while ensuring that the asset-class weights for each investor sum to 100% as showen in Table 13.

Table 13: Target asset-class weights by investor scenario

Asset class	Risky	Balanced	Conservative
Equity	30.96%	18.37%	10.09%
Fixed income	12.48%	26.02%	33.21%
Cash	6.32%	23.09%	34.31%
Real estate	26.26%	17.40%	11.56%
Commodity	23.99%	15.12%	10.83%
Total	100%	100%	100%

We can see patterns arrise based on the expected return for each of the asset class. Fixed Income and Cash assets tend to increase monotonically from risky to conservative profiles, while Equity, Real estate, and Commodity move in oopposite direction.

8.4 Phase III results: common market-data estimates

The market-data inputs are held constant. The downloaded sample extends from 7-years for the whole dataset from October 2019 to October 2026. After calculating euro-denominated daily returns, the in-sample period contains 1,257 common observations for period of 5-years. The out-of-sample period contains 501 price observations from 2-years.

Asset universe has been defined capturing 119 instruments belong to the five modelled asset classes and possess the data required by the optimiser. Their in-sample annualised returns, volatilities, and covariance matrix are used unchanged for all three investors.

8.5 Phase IV results: portfolio construction and optimisation

8.5.1 Risky investor portfolio

The risky portfolio selects nine ETFs. The largest positions are BNO, SMH, SOXX, and XLRE at the 15% security cap, followed by JNK at 12.48%, VNQ at 11.26%, SLV at 8.99%, BIL at 6.32%, and ARGT at 0.96%. The allocation therefore uses commodities, semiconductor equities, listed real estate, and high-yield credit to target growth. It has an in-sample expected annual return of 17.59%, volatility of 21.66%, variance of 0.0469, and Sharpe ratio of 0.812.

8.5.2 Balanced investor portfolio

The balanced portfolio selects eleven ETFs. BIL receives 23.09%, while SMH, XLRE, and SCHP each receive 15%. The remaining allocation is distributed across VGSH (8.56%), IAU (6.53%), DBA (5.90%), ARGT (3.37%), BNO (2.69%), TIP (2.45%), and VNQ (2.40%). The combination of Treasury bills, inflation-linked bonds, equities, real estate, and commodities reflects the investor's mixed growth and preserve capital objective. Expected annual return is 11.01%, volatility is 12.72%, variance is 0.0162, and the Sharpe ratio is 0.865.

8.5.3 Conservative investor portfolio

The conservative portfolio selects ten ETFs and assigns 34.31% to BIL. SCHP and VGSH each receive 15%, followed by XLRE at 11.56%, SMH at 10.09%, DBA at 5.58%, IAU at 3.89%, VCSH at 3.21%, BNO at 1.29%, and CPER at 0.06%. Cash and short-duration or inflation-linked fixed income dominate the allocation, although smaller growth and real-asset exposures maintain diversification across. Expected annual return is 7.54%, volatility is 9.80%, variance is 0.0096, and the Sharpe ratio is 0.769.

Table 14 summarise the expected annual return, volatile, variance, sharp ratio and number of holdings.

Table 14: Optimised in-sample portfolio characteristics by scenario

Metric	Risky	Balanced	Conservative
Expected annual return	17.59%	11.01%	7.54%
Annualised volatility	21.66%	12.72%	9.80%
Variance	0.0469	0.0162	0.0096
Sharpe ratio	0.812	0.865	0.769
Number of holdings	9	11	10

8.6 Phase V results: benchmarking and performance analysis

8.6.1 Out-of-sample performance

We perform out-of-samople for 2-years over the three investor profiles, the risky portfolio outperform SPY, market proxy benchmark. The balanced portfolio tracked SPY but finished slightly below SPY, while the conservative portfolio finished significantly below SPY, All the three portfolios achieve a higher Sharpe ratio than the benchmark, although the comparison is specific to this two-year sample. Figure 15 shows the three portfolios over the 2-years starting with €1 million.

Table 15: Out-of-sample performance by investor scenario

Metric	Risky	Balanced	Conservative	S&P 500 (EUR)
Total return	70.56%	32.71%	22.53%	36.10%
Annualised return	30.67%	15.23%	10.72%	16.70%
Annualised volatility	21.03%	12.22%	9.79%	17.88%
Sharpe ratio	1.458	1.247	1.095	0.934
Maximum drawdown	20.87%	13.08%	12.09%	23.03%
Ending value	€1,705,595	€1,327,058	€1,225,349	€1,360,970

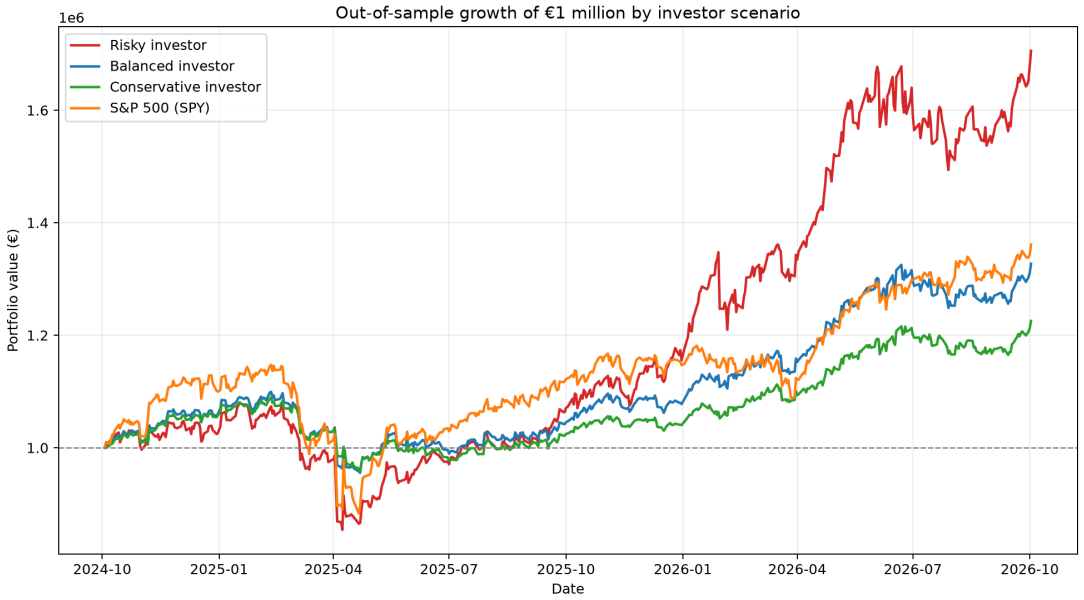


Figure 6: Growth of €1 million: three optimised investor portfolios versus the S&P 500

8.6.2 Downside-risk results

At the 95% confidence level, the loss thresholds on €1 million are approximately €21,864, €12,359, and €9,241 for the risky, balanced, and conservative portfolios, respectively. Corresponding Expected Shortfalls are approximately €33,160, €17,696, and €13,740.

As expected, the risky portfolio has greater tail exposure than SPY, whereas both the balanced and conservative portfolios reduce VaR and ES at both confidence levels. Table 16 showes out-of-sample historical one-day VaR and ES.

Table 16: Out-of-sample historical one-day VaR and Expected Shortfall

Strategy	95% VaR	95% ES	99% VaR	99% ES
Risky	2.19%	3.32%	3.86%	5.85%
Balanced	1.24%	1.77%	2.35%	2.97%
Conservative	0.92%	1.37%	2.12%	2.31%
S&P 500 (EUR)	1.64%	2.63%	3.30%	4.73%

8.6.3 Monte Carlo simulation results

Finally, applying Monte Carlo simulation, the risky portfolio has the highest median wealth, but its 5th percentile is lower than those of the balanced and conservative portfolios. The conservative portfolio has the highest 5th-percentile wealth, indicating stronger downside protection, while retaining the lowest upside, as shown in Table 17.

Table 17: One-year Monte Carlo terminal-wealth results

Strategy	Median	5th percentile	95th percentile
Risky	€1,161,882	€820,441	€1,657,255
Balanced	€1,108,812	€897,039	€1,364,307
Conservative	€1,073,354	€912,024	€1,259,734
S&P 500 (EUR)	€1,160,958	€812,040	€1,666,150

9 Conclusion

In thie report, we discussed an end-to-end framework for constructing and evaluating portfolio based on set of investor's key attributes. The framework link investor preference to portfolio decisions through five phases: investor risk assessment, asset-class suitability, market-data preparation, security-level optimisation, and out-of-sample performance evaluation. Three investor scenarios were examined as a broad senstivity analysis for the in-sample dataset, The risky investor received the highest allocation to equities, real estate, and commodities, producing the highest expected return but also the greatest volatility and tail risk. The conservative investor received the largest allocation to cash and fixed income, resulting in the lowest volatility, drawdown, VaR, and Expected Shortfall. The balanced portfolio occupied the middle ground and provided a more even compromise between growth and capital preservation.

A Full Asset Universe

This appendix reproduces the complete asset universe from `asset_universe.csv`.

Table 18: Complete investable asset universe

Ticker	Display name	Type	Asset class	Asset subclass	Geography
SPY	SPDR S&P 500 ETF Trust	etf	Equity	US large blend	United States
IVV	iShares Core S&P 500 ETF	etf	Equity	US large blend	United States
VOO	Vanguard S&P 500 ETF	etf	Equity	US large blend	United States
VTI	Vanguard Total Stock Market ETF	etf	Equity	US total market	United States
RSP	Invesco S&P 500 Equal Weight ETF	etf	Equity	US large equal weight	United States
QQQ	Invesco QQQ Trust	etf	Equity	US large growth	United States
DIA	SPDR Dow Jones Industrial Average ETF	etf	Equity	US large blend	United States
IWM	iShares Russell 2000 ETF	etf	Equity	US small blend	United States
VTWO	Vanguard Russell 2000 ETF	etf	Equity	US small blend	United States
IWB	iShares Russell 1000 ETF	etf	Equity	US large blend	United States
IWF	iShares Russell 1000 Growth ETF	etf	Equity	US large growth	United States
IWD	iShares Russell 1000 Value ETF	etf	Equity	US large value	United States
VUG	Vanguard Growth ETF	etf	Equity	US large growth	United States
VTV	Vanguard Value ETF	etf	Equity	US large value	United States
SCHB	Schwab US Broad Market ETF	etf	Equity	US broad market	United States
SCHG	Schwab US Large-Cap Growth ETF	etf	Equity	US large growth	United States
SCHV	Schwab US Large-Cap Value ETF	etf	Equity	US large value	United States
QUAL	iShares MSCI USA Quality Factor ETF	etf	Equity	US quality factor	United States
USMV	iShares MSCI USA Min Vol Factor ETF	etf	Equity	US minimum volatility	United States
MTUM	iShares MSCI USA Momentum Factor ETF	etf	Equity	US momentum factor	United States
VLUE	iShares MSCI USA Value Factor ETF	etf	Equity	US value factor	United States
SIZE	iShares MSCI USA Size Factor ETF	etf	Equity	US size factor	United States
VIG	Vanguard Dividend Appreciation ETF	etf	Equity	US dividend growth	United States
SCHD	Schwab US Dividend Equity ETF	etf	Equity	US dividend equity	United States
XLK	Technology Select Sector SPDR Fund	etf	Equity	Technology sector	United States
XLF	Financial Select Sector SPDR Fund	etf	Equity	Financial sector	United States
XLV	Health Care Select Sector SPDR Fund	etf	Equity	Health care sector	United States
XLY	Consumer Discretionary Select Sector SPDR Fund	etf	Equity	Consumer discretionary sector	United States
XLP	Consumer Staples Select Sector SPDR Fund	etf	Equity	Consumer staples sector	United States
XLE	Energy Select Sector SPDR Fund	etf	Equity	Energy sector	United States
XLI	Industrial Select Sector SPDR Fund	etf	Equity	Industrials sector	United States
XLB	Materials Select Sector SPDR Fund	etf	Equity	Materials sector	United States
XLU	Utilities Select Sector SPDR Fund	etf	Equity	Utilities sector	United States
XLRE	Real Estate Select Sector SPDR Fund	etf	Real estate	Listed real estate	United States
XLC	Communication Services Select Sector SPDR Fund	etf	Equity	Communication services sector	United States
SMH	VanEck Semiconductor ETF	etf	Equity	Semiconductors	Global

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Table 18: Complete investable asset universe (continued)

Ticker	Display name	Type	Asset class	Asset subclass	Geography
SOXX	iShares Semiconductor ETF	etf	Equity	Semiconductors	United States
XBI	SPDR S&P Biotech ETF	etf	Equity	Biotechnology	United States
IBB	iShares Biotechnology ETF	etf	Equity	Biotechnology	United States
ITA	iShares US Aerospace and Defense ETF	etf	Equity	Aerospace and defense	United States
XRT	SPDR S&P Retail ETF	etf	Equity	Retail industry	United States
KRE	SPDR S&P Regional Banking ETF	etf	Equity	Regional banks	United States
IYR	iShares US Real Estate ETF	etf	Real estate	Listed real estate	United States
VNQ	Vanguard Real Estate ETF	etf	Real estate	Listed real estate	United States
TAN	Invesco Solar ETF	etf	Equity	Clean energy solar	Global
ICLN	iShares Global Clean Energy ETF	etf	Equity	Clean energy	Global
BOTZ	Global X Robotics and Artificial Intelligence ETF	etf	Equity	Robotics and AI	Global
ARKK	ARK Innovation ETF	etf	Equity	Disruptive innovation	United States
CIBR	First Trust Nasdaq Cybersecurity ETF	etf	Equity	Cybersecurity	Global
VXUS	Vanguard Total International Stock ETF	etf	Equity	International ex US	Global ex US
VEA	Vanguard FTSE Developed Markets ETF	etf	Equity	Developed markets	Developed ex US
VWO	Vanguard FTSE Emerging Markets ETF	etf	Equity	Emerging markets	Emerging markets
IEFA	iShares Core MSCI EAFE ETF	etf	Equity	Developed markets	Developed ex US
IEMG	iShares Core MSCI Emerging Markets ETF	etf	Equity	Emerging markets	Emerging markets
EFA	iShares MSCI EAFE ETF	etf	Equity	Developed markets	Developed ex US
EEM	iShares MSCI Emerging Markets ETF	etf	Equity	Emerging markets	Emerging markets
ACWI	iShares MSCI ACWI ETF	etf	Equity	Global equity	Global
VT	Vanguard Total World Stock ETF	etf	Equity	Global equity	Global
EWJ	iShares MSCI Japan ETF	etf	Equity	Country equity	Japan
EWZ	iShares MSCI Brazil ETF	etf	Equity	Country equity	Brazil
INDA	iShares MSCI India ETF	etf	Equity	Country equity	India
MCHI	iShares MSCI China ETF	etf	Equity	Country equity	China
FXI	iShares China Large-Cap ETF	etf	Equity	Country equity	China
EWG	iShares MSCI Germany ETF	etf	Equity	Country equity	Germany
EWU	iShares MSCI United Kingdom ETF	etf	Equity	Country equity	United Kingdom
EWQ	iShares MSCI France ETF	etf	Equity	Country equity	France
EWI	iShares MSCI Italy ETF	etf	Equity	Country equity	Italy
EWP	iShares MSCI Spain ETF	etf	Equity	Country equity	Spain
EWC	iShares MSCI Canada ETF	etf	Equity	Country equity	Canada
EWA	iShares MSCI Australia ETF	etf	Equity	Country equity	Australia
EWH	iShares MSCI Hong Kong ETF	etf	Equity	Country equity	Hong Kong
EWS	iShares MSCI Singapore ETF	etf	Equity	Country equity	Singapore
EWT	iShares MSCI Taiwan ETF	etf	Equity	Country equity	Taiwan
EWY	iShares MSCI South Korea ETF	etf	Equity	Country equity	South Korea
EWW	iShares MSCI Mexico ETF	etf	Equity	Country equity	Mexico
EZA	iShares MSCI South Africa ETF	etf	Equity	Country equity	South Africa
TUR	iShares MSCI Turkey ETF	etf	Equity	Country equity	Turkey
THD	iShares MSCI Thailand ETF	etf	Equity	Country equity	Thailand

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Table 18: Complete investable asset universe (continued)

Ticker	Display name	Type	Asset class	Asset subclass	Geography
EIDO	iShares MSCI Indonesia ETF	etf	Equity	Country equity	Indonesia
EPOL	iShares MSCI Poland ETF	etf	Equity	Country equity	Poland
GREK	Global X MSCI Greece ETF	etf	Equity	Country equity	Greece
ARGT	Global X MSCI Argentina ETF	etf	Equity	Country equity	Argentina
KSA	iShares MSCI Saudi Arabia ETF	etf	Equity	Country equity	Saudi Arabia
UAE	iShares MSCI UAE ETF	etf	Equity	Country equity	United Arab Emirates
AGG	iShares Core US Aggregate Bond ETF	etf	Fixed income	Aggregate bonds	United States
BND	Vanguard Total Bond Market ETF	etf	Fixed income	Aggregate bonds	United States
BNDX	Vanguard Total International Bond ETF	etf	Fixed income	International aggregate bonds	Global ex US
TLT	iShares 20 Plus Year Treasury Bond ETF	etf	Fixed income	Long government bonds	United States
IEF	iShares 7-10 Year Treasury Bond ETF	etf	Fixed income	Intermediate government bonds	United States
SHY	iShares 1-3 Year Treasury Bond ETF	etf	Fixed income	Short government bonds	United States
VGSH	Vanguard Short-Term Treasury ETF	etf	Fixed income	Short government bonds	United States
VGIT	Vanguard Intermediate-Term Treasury ETF	etf	Fixed income	Intermediate government bonds	United States
VGLT	Vanguard Long-Term Treasury ETF	etf	Fixed income	Long government bonds	United States
GOVT	iShares US Treasury Bond ETF	etf	Fixed income	Government bonds	United States
TIP	iShares TIPS Bond ETF	etf	Fixed income	Inflation-linked bonds	United States
SCHP	Schwab US TIPS ETF	etf	Fixed income	Inflation-linked bonds	United States
LQD	iShares iBoxx Investment Grade Corporate Bond ETF	etf	Fixed income	Investment grade corporate	United States
VCIT	Vanguard Intermediate-Term Corporate Bond ETF	etf	Fixed income	Investment grade corporate	United States
VCSH	Vanguard Short-Term Corporate Bond ETF	etf	Fixed income	Investment grade corporate	United States
HYG	iShares iBoxx High Yield Corporate Bond ETF	etf	Fixed income	High yield corporate	United States
JNK	SPDR Bloomberg High Yield Bond ETF	etf	Fixed income	High yield corporate	United States
EMB	iShares JPMorgan USD Emerging Markets Bond ETF	etf	Fixed income	Emerging market sovereign	Emerging markets
MUB	iShares National Muni Bond ETF	etf	Fixed income	Municipal bonds	United States
BIL	SPDR Bloomberg 1-3 Month T-Bill ETF	etf	Cash	Treasury bills	United States
GLD	SPDR Gold Shares	etf	Commodity	Gold	Global
IAU	iShares Gold Trust	etf	Commodity	Gold	Global
SLV	iShares Silver Trust	etf	Commodity	Silver	Global
PPLT	abrdn Physical Platinum Shares ETF	etf	Commodity	Platinum	Global
PALL	abrdn Physical Palladium Shares ETF	etf	Commodity	Palladium	Global
DBC	Invesco DB Commodity Index Tracking Fund	etf	Commodity	Broad commodities	Global
PDBC	Invesco Optimum Yield Diversified Commodity Strategy ETF	etf	Commodity	Broad commodities	Global
GSG	iShares S&P GSCI Commodity-Indexed Trust	etf	Commodity	Broad commodities	Global
USO	United States Oil Fund	etf	Commodity	Crude oil	Global
BNO	United States Brent Oil Fund	etf	Commodity	Brent crude oil	Global
UNG	United States Natural Gas Fund	etf	Commodity	Natural gas	Global
DBA	Invesco DB Agriculture Fund	etf	Commodity	Agriculture	Global
CORN	Teucrium Corn Fund	etf	Commodity	Corn	Global

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Table 18: Complete investable asset universe (continued)

Ticker	Display name	Type	Asset class	Asset subclass	Geography
WEAT	Teucrium Wheat Fund	etf	Commodity	Wheat	Global
CPER	United States Copper Index Fund	etf	Commodity	Copper	Global
GBTC	Grayscale Bitcoin Trust ETF	etf	Crypto asset	Spot Bitcoin	Global
ETHE	Grayscale Ethereum Trust ETF	etf	Crypto asset	Spot Ether	Global
VXX	iPath Series B S&P 500 VIX Short-Term Futures ETN	etn	Volatility	Long volatility	United States
UVXY	ProShares Ultra VIX Short-Term Futures ETF	etf	Volatility	Leveraged long volatility	United States
SVXY	ProShares Short VIX Short-Term Futures ETF	etf	Volatility	Short volatility	United States
VIXY	ProShares VIX Short-Term Futures ETF	etf	Volatility	Long volatility	United States
^VIX	CBOE Volatility Index	index	Volatility	Implied volatility index	United States
UUP	Invesco DB US Dollar Index Bullish Fund	etf	Currency	US dollar	United States
FXE	Invesco CurrencyShares Euro Trust	etf	Currency	Euro	Eurozone
FXY	Invesco CurrencyShares Japanese Yen Trust	etf	Currency	Japanese yen	Japan
FXB	Invesco CurrencyShares British Pound Sterling Trust	etf	Currency	British pound	United Kingdom
FXA	Invesco CurrencyShares Australian Dollar Trust	etf	Currency	Australian dollar	Australia
FXC	Invesco CurrencyShares Canadian Dollar Trust	etf	Currency	Canadian dollar	Canada
CEW	WisdomTree Emerging Currency Strategy Fund	etf	Currency	Emerging market currencies	Emerging markets
DBMF	iMGP DBi Managed Futures Strategy ETF	etf	Alternatives	Managed futures	Global

B Reproducibility

- Notebook: `FIN42350 Portfolio Design Project.ipynb`
- Asset universe: `data/asset_universe.csv`
- Dashboard application: `source/portfolio.py`